

Classes

Classes composition:

Input File Structure of the input file:

Mesh points: Number of points in X, Y, Z directions

Example: 50 50 50 $\Rightarrow Nx = 50, Ny = 50, Nz = 50$

Time parameters:

dt - Time step size, e.g., 0.01

T - Total simulation time, e.g., 10.0

Spatial step sizes:

dx dy dz - Grid spacing in X, Y, Z, e.g., 0.5 0.5 0.5

Initial conditions files:

u0_init_file - File with initial velocity values

p0_init_file - File with initial pressure values

k_file - File with permeability values

Boundary conditions analysis

shrink Velocity Boundary Conditions For each of the three steps of the algorithm for momentum equation, the Dir when solving for the component parallel to the decomposition direction:

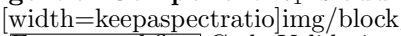
when solving for a component perpendicular to the decomposition direction:

Boundary Conditions: Indexing Pattern

We found a structural pattern in our staggered grid that simplifies boundary condition (BC) logic. Let x_0 be the s
0.6

Normal Component: i_1 is **even** AND i_2 is **even**.

Tangential Component: i_1 is **odd** OR i_2 is **odd**.

0.4  This pattern holds for all **three velocity components** and allows for

Future workflow Code Validation

Compute error

Compute residual

Convergence study

Parallelization Strategy

Vectorization of the sequential code

Parallelization of the code